www.tntextbooks.in

7.

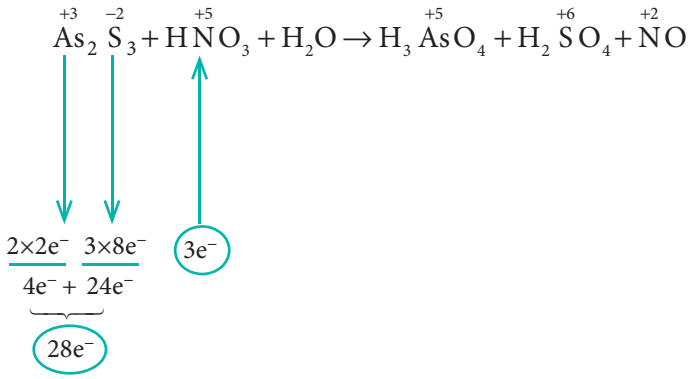
Content	Reactant		Products	
	x	y	*l*	m
Stoichiometric coefficient	2	3	4	1
No. of moles allowed to react	8	15	–	–
No. of moles of reactant reacted and product formed	8	12	16	4
No. of moles of un-reacted reactants and the product formed	–	3	16	4

Limiting reagent : **x**

Product formed : **16 moles of *l* & 4 moles of m**

Amount of excess reactant : **3 moles of y**

8.



Equate the total no. of electrons in the reactant side by cross multiplying,

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 + H_2O \rightarrow H_3AsO_4 + H_2SO_4 + NO$

Based on reactant side, balance the products

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 + H_2O \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$

Product side :36 hydrogen atoms & 88 oxygen atoms

Reactant side :28 hydrogen atoms & 74 oxygen atoms

Difference is 8 hydrogen atoms & 4 oxygen atoms

$\therefore$ Multiply H_2O molecule on the reactant side by '4'.

Balanced equation is,

$3\ As_2S_3 + 28\ HNO_3 + 4\ H_2O \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$